Introduction to Homotopy Type Theory

HOMOTOPY PULLBACKS

2024-01-09

Reference: [Rijke 2018, Introduction to HoTT, Chapter 10]

<u>Goals</u>

- Define pullbacks using the universal property and characterize the explicit higher data which determines if a commutative square is a pullback.
- Show that pullbacks are uniquely unique
- Given a cospan $f : A \to X \leftarrow B : g$, define a canonical pullback type $A \times_X B$, show that for any other cone (p, q, H) with vertex C, there is a map

$$\mathrm{gap}(p,q,H) : C \to A \times_X B$$

such that (p,q,H) is a pullback $\iff$ $\mathrm{gap}(p,q,H)$ is an equiv.

- Some families of examples:
  - Products
  - Fiber products
  - Fibers
  - Fiber sequences
- Some extra applications:
  - Characterization of fiberwise equivalences
  - Preservation of truncation and equivalences
  - Pullback pasting
  - Disjointness of coproducts

Pullback Squares

A square
$$\begin{tikzcd} C \arrow[r, "q"] \arrow[d, "p"'] & B \arrow[d, "g"] \\ A \arrow[r, "f"'] & X \end{tikzcd}$$
is said to **commute** if there is a homotopy $H : f\circ p \sim g \circ q$ $\left(\equiv \prod_{c:C} f\circ p(c) =_X g\circ p(c) \right)$

$$\left\{ \begin{array}{c} \text{Commutative squares} \\ \begin{tikzcd} C \arrow[r, "q"] \arrow[d, "p"'] \arrow[dr, phantom, "H \Nearrow"] & B \arrow[d, "g"] \\ A \arrow[r, "f"'] & X \end{tikzcd} \end{array} \right\} \equiv \left\{ \begin{array}{c} \text{Cones over a cospan} \\ A \xrightarrow[f]{} X \xleftarrow[g]{} B \\ \text{with vertex } C \end{array} \right\} \equiv \mathrm{cone}(C)$$

where $\mathrm{cone}(C) :\equiv \sum_{p:C\to A} \sum_{q:C\to B} f\circ p \sim g \circ q$

For any commuting square as before and any type C', define:

$$\begin{aligned} \mathrm{coneMap}(p,q,H) : (C' \to C) &\longrightarrow \mathrm{cone}(C') \\ h &\longmapsto (p\circ h, q\circ h, H\circ h) \end{aligned}$$

$$\begin{tikzcd} C' \arrow[dr, "h"'] & & \\ & C \arrow[r, "q"] \arrow[d, "p"'] & B \arrow[d, "g"] \\ & A \arrow[r, "f"'] & X \end{tikzcd} \qquad \begin{tikzcd} C' \arrow[dr, "h"'] \arrow[drr, dashed, bend left, "q\circ h"] \arrow[ddr, dashed, bend right, "p\circ h"'] & & \\ & C \arrow[r, "q"] \arrow[d, "p"'] & B \arrow[d, "g"] \\ & A \arrow[r, "f"'] & X \end{tikzcd}$$

A commuting square as before is a **pullback square** if $\mathrm{coneMap}(p,q,H)$ is an equivalence for any type C'.

Corresponds to the universal property of the pullback

$$\begin{tikzcd} C' \arrow[dr, "h"'] \arrow[drr, bend left, "q'"] \arrow[ddr, bend right, "p"'] & & \\ & C \arrow[r, "q"] \arrow[d, "p"'] & B \arrow[d, "g"] \\ & A \arrow[r, "f"'] & X \end{tikzcd}$$

To understand the universal property, first we need to characterize the identity type of $\mathrm{cone}(C)$:

Lemma 1 Let (p,q,H) and (p',q',H') be cones on a cospan $f : A \longrightarrow X \leftarrow B : g$ and vertex C. Then
$$\big((p,q,H) = (p',q',H')\big) \simeq (K, L, M) \quad \text{where:}$$
$K : p \sim p'$, $L : q \sim q'$ and $M : H \cdot (g\circ L) \sim (f\circ K)\cdot H'$

$$\begin{tikzcd} C \arrow[dr, equal] \arrow[drr, bend left, "q'"] \arrow[ddr, bend right, "p'"'] & & \\ & C \arrow[r, "q"] \arrow[d, "p"'] \arrow[ur, phantom, "\Downarrow L"] \arrow[dl, phantom, "\overset{K}{\Rightarrow}"] & B \arrow[d, "g"] \\ & A \arrow[r, "f"'] & X \end{tikzcd} \qquad \begin{tikzcd} f\circ p \arrow[r, "f\circ K", "\sim"'] \arrow[d, "H"', "\wr"] \arrow[dr, phantom, "M \Nearrow"] & f\circ p' \arrow[d, "H'", "\wr"'] \\ g\circ q \arrow[r, "\sim", "g\circ L"'] & g\circ q' \end{tikzcd}$$

Proof How we characterize identity type of a type A?

1. Define a a candidate $\prod_{x,y:A} R(x,y)$
2. Prove that R is reflective $\prod_{x:A} R(x,x)$ obtaining a fiberwise map by path induction:
$$\prod_{x,y:A} (x=y) \to R(x,y) \qquad (*)$$
3. Prove that $\sum_{x:A} R(a,x)$ is contractible, and then by the fundamental theorem of identity types, $(*)$ is an equivalence.

In this case,

(1) $R((p,q,H),(p',q',H')) = (K,L,M)$

(2) $(\mathrm{htpyRefl}_p, \mathrm{htpyRefl}_q, \mathrm{htpyRefl}_H) : R((p,q,H),(p,q,H))$

By (3) and associativity of Σ, it suffices to show that
$$\sum_{p'} \sum_{q'} \sum_{H'} \sum_{K:p\sim p'} \sum_{L:q\sim q'} H\cdot (g\circ L) \sim (f\circ K)\cdot H' \simeq$$
is contractible.

Swap the order of q', H', K and L + Σ associativity

$$\simeq \sum_{\underbrace{\left((p',K) : \sum_{p'} p \sim p'\right)}} \; \sum_{\underbrace{\left((q',L) : \sum_{q'} q\sim q'\right)}} \; \sum_{H'} H\cdot(g\circ L) \sim (f\circ K)\cdot H' \simeq$$

Both are contractible by function extensionality with centers of contraction $(p, \mathrm{htpyRefl}_p)$ and $(q, \mathrm{htpyRefl}\, q)$

$\left(\text{Exercise: } C \text{ contractible with center } c \Rightarrow \sum_{x:C} B(x) \simeq B(c) \right)$

$$\simeq \sum_{H'} H \cdot \mathrm{htpyRefl}_{g\circ q} \sim \mathrm{htpyRefl}_{f\circ p} \cdot H' \simeq \sum_{H'} H \sim H'$$

Also contractible by fun. ext. ∎

Corollary 2
$$\begin{tikzcd} C \arrow[r, "q"] \arrow[d, "p"'] \arrow[dr, phantom, "H \Nearrow"] & B \arrow[d, "g"] \\ A \arrow[r, "f"'] & X \end{tikzcd}$$
is a pullback

$\Updownarrow$

For every type C' and cone (p',q',H') with vertex C' the types of quadruples (h, K, L, M) consisting of
$$h : C' \to C \qquad \begin{aligned} K &: p\circ h \sim p' \\ L &: q\circ h \sim q' \end{aligned} \qquad M : (H\circ h)\cdot (g\circ L) \sim (f\circ K)\cdot H'$$
is contractible.

$$\begin{tikzcd} C' \arrow[dr, "h"] \arrow[drr, bend left, "q'"] \arrow[ddr, bend right, "p'"'] & & \\ & C \arrow[r, "q"] \arrow[d, "p"'] \arrow[ur, phantom, "\Downarrow L"] \arrow[dl, phantom, "\overset{K}{\Rightarrow}"] & B \arrow[d, "g"] \\ & A \arrow[r, "f"'] & X \end{tikzcd} \qquad \begin{tikzcd} f\circ p \arrow[r, "f\circ K", "\sim"'] \arrow[d, "H\circ h"', "\wr"] \arrow[dr, phantom, "M \Nearrow"] & f\circ p' \arrow[d, "H'", "\wr"'] \\ g\circ q \arrow[r, "\sim", "g\circ L"'] & g\circ q' \end{tikzcd}$$

Uniqueness of pullbacks

The uniqueness of pullbacks up to equivalence follows from:

Lemma 3 Suppose

$$
\begin{tikzcd}
C \arrow[r, "q"] \arrow[d, "p"'] \arrow[dr, phantom, "H \Nearrow"] & B \arrow[d, "g"] \\
A \arrow[r, "f"'] & X
\end{tikzcd}
\qquad
\begin{tikzcd}
C' \arrow[r, "q'"] \arrow[d, "p'"'] \arrow[dr, phantom, "H' \Nearrow"] & B \arrow[d, "g"] \\
A \arrow[r, "f"'] & X
\end{tikzcd}
$$

$$h : C \to C'$$
$$K : p \circ h \sim p' \quad L : q \circ h \sim q'$$
$$M : (H \circ h) \bullet (g \circ L) \sim (f \circ K)$$

Then, if any two of the following hold, so does the third:

(i) C is a pullback
(ii) C' is a pullback
(iii) h is an equivalence

Proof By the characterization of the identity types of $\mathrm{cone}(C')$, from (K, L, M) we obtain

$$\mathrm{coneMap}((p, q, H), h) = (p', q', H') \qquad (1)$$

For any type D and $k : D \to C'$ a map,

$$
\begin{aligned}
\mathrm{coneMap}((p, q, H), (h \circ k)) &\equiv (p \circ h \circ k,\ q \circ h \circ k,\ H \circ h \circ k) \\
&\equiv \mathrm{coneMap}(\mathrm{coneMap}((p, q, H), h), k) \\
&\underset{(1)}{=} \mathrm{coneMap}((p', q', H'), k)
\end{aligned}
$$

Thus, there is a commutative triangle:

$$
\begin{tikzcd}
(D \to C') \arrow[rr, "h \circ -"] \arrow[dr, "\mathrm{coneMap}(p', q', H')"'] & & (D \to C) \arrow[dl, "\mathrm{coneMap}(p, q, H)"] \\
& \mathrm{cone}(D) &
\end{tikzcd}
$$

Then, the result follows from the 3-for-2 property of equiv. and $h \circ -$ is an equivalence for any $D \Leftrightarrow h$ is an equivalence $\square$

As a corollary, we obtain that the pullbacks are not only unique, but "uniquely unique":

Corollary 4 Consider two pullback squares

$$
\begin{tikzcd}
C \arrow[r, "q"] \arrow[d, "p"'] \arrow[dr, phantom, "H \Nearrow"] & B \arrow[d, "g"] \\
A \arrow[r, "f"'] & X
\end{tikzcd}
\qquad
\begin{tikzcd}
C' \arrow[r, "q'"] \arrow[d, "p'"'] \arrow[dr, phantom, "H' \Nearrow"] & B \arrow[d, "g"] \\
A \arrow[r, "f"'] & X
\end{tikzcd}
$$

Then the types of quadruples (e, K, L, M) consisting of

$$e : C' \simeq C \qquad \begin{matrix} K : p \circ e \sim p' \\ L : q \circ e \sim q' \end{matrix} \qquad M : (H \circ e) \bullet (g \circ L) \sim (f \circ K) \bullet H'$$

is contractible.

Proof By corollary 2, the type of quadruples (h, K, L, M), with $h : C' \to C$ being a map, is contractible.

By lemma 3, h is an equivalence. Then, because $\mathrm{isEquiv}(h)$ is a non-empty proposition, it is contractible.

The type that we want to study is equivalent to

$$\sum_{(h, K, L, M)} \mathrm{isEquiv}(h) \qquad (\text{by } \Sigma\text{-swap})$$

which is a dependent product of a contractible type and a contractible family, thus contractible $\square$

Canonical pullbacks

Let $f : A \to X \leftarrow B : g$ a cospan. Define the canonical pullback $A \times_X B$ of f and g as

$$A \times_X B :\equiv \sum_{x:A} \sum_{y:B} f(x) = g(y)$$

$$\pi_1 :\equiv pr_1 : A \times_X B \to A \qquad \pi_2 :\equiv pr_1 \circ pr_2 : A \times_X B \to B$$

$$\pi_3 :\equiv pr_2 \circ pr_2 : f \circ \pi_1 \sim g \circ \pi_2$$

$$\begin{tikzcd} A \times_X B \arrow[r, "\pi_2"] \arrow[d, "\pi_1"'] \arrow[dr, phantom, "\pi_3 \swarrow"] & B \arrow[d, "g"] \\ A \arrow[r, "f"'] & X \end{tikzcd} \text{ is a pullback}$$

Proof $(C \to A \times_X B) \equiv C \to \sum_{x:A} \sum_{y:B} f(x) = g(x)$

$$\simeq \sum_{p:C\to A} \prod_{z:C} \sum_{y:B} f(p(z)) = y$$

$$\simeq \sum_{p:C\to A} \sum_{q:C\to B} \prod_{z:C} f(p(z)) = g(q(z))$$

$$\equiv cone(C)$$

By definition this map is $coneMap(\pi_1, \pi_2, \pi_3)$.

$$\left(\prod_{x:A} \sum_{y:B(x)} C(x,y) \simeq \sum_{(f:\prod_{(x:A)} B(x))} \prod_{x:A} C(x, f(x)) \right.$$
$$\left. h \longmapsto (\lambda x. pr_1(h(x)), \lambda x. pr_2(h(x))) \right)$$ ∎

Given a commutative square $\begin{tikzcd} C \arrow[r, "q"] \arrow[d, "p"'] \arrow[dr, phantom, "H \swarrow"] & B \arrow[d, "g"] \\ A \arrow[r, "f"'] & X \end{tikzcd}$ define the gap map

$$gap(p,q,H) : C \longrightarrow A \times_X B$$
$$z \longmapsto (p(z), q(z), H(z))$$

and define $isPullback(p,q,H) :\equiv isEquiv(gap(p,q,H))$

Theorem 5 $\begin{tikzcd} C \arrow[r, "q"] \arrow[d, "p"'] \arrow[dr, phantom, "H \swarrow"] & B \arrow[d, "g"] \\ A \arrow[r, "f"'] & X \end{tikzcd}$ is a pullback $\Leftrightarrow$ There is a term of $isPullback(p,q,H)$

Proof By lemma 3, it suffices to show that there exist

$K : \pi_1 \circ gap(p,q,H) \sim p \qquad L : \pi_2 \circ gap(p,q,H) \sim q$

$K :\equiv \lambda z. refl_{p(z)} \qquad L :\equiv \lambda z. refl_{q(z)}$

$$M : (\pi_3 \circ gap(p,q,H)) \cdot (g \circ L) \sim (f \circ K) \cdot H$$

$$M :\equiv \lambda z. rightUnit(H(z)) \cdot (leftUnit(H(z)))^{-1}$$ ∎

$$M(z) : H(z) \cdot \underbrace{(ap_g(refl_{q(z)}))}_{refl_{g(q(z))}} = \underbrace{(ap_f(refl_{p(z)}))}_{refl_{f(p(z))}} \cdot H(z)$$

$$\begin{tikzcd} C \arrow[dr, "gap(p,q,H)"] \arrow[drr, bend left, "q"] \arrow[ddr, bend right, "p"'] & & \\ & A \times_X B \arrow[r, "\pi_2"] \arrow[d, "\pi_1"] & B \arrow[d, "g"] \\ & A \arrow[r, "f"'] & X \end{tikzcd}$$

Fiber products and fibers as pullbacks

Product:

The square
$$\begin{tikzcd} A\times B \arrow[r, "pr_2"] \arrow[d, "pr_1"'] & B \arrow[d, "const_*"] \\ A \arrow[r, "const_*"'] & 1 \end{tikzcd}$$
commuting by $const_{refl_*}$ is a pullback

Proof By theorem 5, it suffices to show that
$$gap(pr_1, pr_2, const_{refl_*})$$
is an equivalence. The inverse is $\lambda(a,b,p).(a,b)$ ◻

Pullbacks as fiber products

Theorem 6 Let P and Q be families over X. Then the square
$$\begin{tikzcd} \sum_{x:X} P(x)\times Q(x) \arrow[r, "\lambda(x,(p,q)).(x,q)"] \arrow[d, "\lambda(x,(p,q)).(x,p)"'] & \sum_{x:X} Q(x) \arrow[d, "pr_1"] \\ \sum_{x:X} P(x) \arrow[r, "pr_1"'] & X \end{tikzcd}$$
commuting by $H :\equiv \lambda(x,(p,q)).refl_x$

is a pullback square.

Proof By theorem 5, it suffices to show that the gap map is an equivalence. The gap map by definition is
$$\lambda(x,(p,q)).((x,p),(x,q),refl_x)$$
The inverse of this map is:
$$\lambda((x,p),(y,q),\alpha).(y,(tr_P(\alpha,p),q))$$
◻

The fiber of $f: A\to B$ at $b:B$ is the type
$$fib_f(b) :\equiv \sum_{a:A} f(a)=b$$

Like the homotopy fiber or the preimage of a function.

For any map $f: A \to B$, there is an equivalence
$$e : A \simeq \sum_{y:B} fib_f(y) \qquad (*)$$
and a commutative triangle
$$\begin{tikzcd} A \arrow[rr, "e", "\sim"'] \arrow[dr, "f"'] & & \sum_{y:B} fib_f(y) \arrow[dl, "pr_1"] \\ & B & \end{tikzcd}$$

Corollary 7 For any cospan $f: A\to X \leftarrow B: g$, the square
$$\begin{tikzcd} \sum_{x:X} fib_f(x)\times fib_g(x) \arrow[r, "\beta"] \arrow[d, "\alpha"'] & B \arrow[d, "g"] \\ A \arrow[r, "f"] & X \end{tikzcd}$$
is a pullback.

where $\alpha :\equiv \lambda(x,((a,p),(b,q))).a$ and $\beta :\equiv \lambda(x,((a,p),(b,q))).b$

Proof By $(*)$, we can expand the square with the following commutative diagram:
$$\begin{tikzcd} & & \sum_{x:X} fib_g(x) \arrow[d, "e^{-1}", "\wr"'] \arrow[dd, bend left=60, "pr_1"] \\ & \sum_{x:X} fib_f(x)\times fib_g(x) \arrow[ur, dashed, "e\circ\beta"] \arrow[r, "\beta"] \arrow[d, "\alpha"] \arrow[dl, dashed, "e\circ\alpha"'] & B \arrow[d, "g"] \\ \sum_{x:X} fib_f(x) \arrow[r, "e^{-1}", "\sim"'] \arrow[rr, bend right, "pr_1"'] & A \arrow[r, "f"] & X \end{tikzcd}$$
By theorem 6, the outer square is a pullback. Finally, composing homotopies we obtain the desired result ◻

Fibers: The square
$$\begin{tikzcd} fib_f(b) \arrow[r, "pr_1"] \arrow[d, "const_*"'] & A \arrow[d, "f"] \\ 1 \arrow[r, "const_b"'] & B \end{tikzcd}$$
commuting by $pr_2 : \prod_{t:fib_f(b)} f(pr_1(t)) = b$ is a pullback

Proof Use corollary 7 with $B\equiv 1$, $X\equiv B$, $g\equiv const_b$, then we can check that $\sum_{y:B} fib_f(y)\times fib_{const_b}(y) \simeq fib_f(b)$ ◻

The previous pullback motivates de definition of fiber sequences:

A *fiber sequence* $F \hookrightarrow E \twoheadrightarrow B$ consists of three types, called the base space B, the total space E and the fiber F, with base points $x:F$, $y:E$ and $b:B$ and a pullback
$$\begin{tikzcd} F \arrow[r, "i"] \arrow[d, "const_*"'] \arrow[dr, phantom, "\lrcorner", very near start] & E \arrow[d, "p"] \\ 1 \arrow[r, "b"'] & B \end{tikzcd}$$
such that $i(x)=y$ and $p(y)=b$.

Example For any type family B over A and any $a:A$,

the square
$$\begin{tikzcd} B(a) \arrow[r, "const_*"] \arrow[d, "\lambda y.(a,y)"'] \arrow[dr, phantom, "\lrcorner", very near start] & 1 \arrow[d, "a"] \\ \sum_{x:A} B(x) \arrow[r, "pr_1"'] & A \end{tikzcd}$$
is a pullback

(The gap map is homotopic to $\lambda y.((a,y), refl_a)$, which is an equivalence between $B(a)$ and $fib_{pr_1}(a)$.)

If there is a term $b: B(a)$, then we obtain a fiber sequence
$$B(a) \hookrightarrow \sum_{x:A} B(x) \longrightarrow A$$

Applications

Characterization of fiberwise equivalences

Consider a fiberwise map $g : \prod_{x:A} P(x) \longrightarrow Q(x)$.

There is a map $\quad total(g) : \sum_{x:A} P(x) \longrightarrow \sum_{x:B} Q(x)$

$$(x,p) \longmapsto (x, g(x,p))$$

such that

g is a fiberwise equivalence $\iff$ $total(g)$ is an equivalence

($g(x)$ equivalence for any $x:A$)

This setting can be further generalized: consider $f : A \to B$ and a fiberwise map $g : \prod_{x:A} P(x) \longrightarrow Q(f(x))$, then define

$$total_f(g) : \sum_{x:A} P(x) \longrightarrow \sum_{y:B} Q(y)$$
$$(x,p) \longmapsto (f(x), g(x,p))$$

Then, we can characterize when g is a fiberwise equivalence:

Theorem Let $f : A \to B$ and $g : \prod_{x:A} P(x) \longrightarrow Q(f(x))$. Then

g is a fiberwise equivalence $\iff$
$$\begin{tikzcd} \sum_{x:A} P(x) \arrow[r, "total_f(g)"] \arrow[d] & \sum_{y:B} Q(y) \arrow[d] \\ A \arrow[r, "f"] & B \end{tikzcd}$$
is a pullback

Preservation of properties

A map $f : A \to B$ is *k-truncated* if for each $y : B$ the fiber $fib_f(b)$ is a k-truncated.

(-2)-truncated maps $\equiv$ fibers contractible $\rightsquigarrow$ Equivalences

(-1)-truncated maps $\equiv$ fibers propositions $\rightsquigarrow$ Embeddings

($f : A \to B$ is an *embedding* if $ap_f : (x = y) \to (f(x) = f(y))$ is an equivalence)

In general, f is $(k+1)$-truncated $\iff$ ap_f is k-truncated

Theorem Consider a pullback square
$$\begin{tikzcd} C \arrow[r, "q"] \arrow[d, "p"'] \arrow[dr, phantom, "\lrcorner", very near start] & B \arrow[d, "g"] \\ A \arrow[r, "f"] & X \end{tikzcd}$$
If g is k-truncated, then so is p

Pullback pasting Consider a commuting diagram

$$\begin{tikzcd} A \arrow[r, "k"] \arrow[d, "f"'] & B \arrow[r, "l"] \arrow[d, "g"] & C \arrow[d, "h"] \\ X \arrow[r, "i"'] & Y \arrow[r, "j"'] & Z \end{tikzcd}$$

with homotopies $H : i \circ f \sim g \circ k$, $K : j \circ g \sim h \circ l$ and
$$(j \circ H) \cdot (K \circ k) : j \circ i \circ f \sim h \circ l \circ k$$

If the right square is a pullback, then

The left square is a pullback $\iff$ The outer square is a pullback

Disjointness of coproducts For any A and B,

the square
$$\begin{tikzcd} 0 \arrow[r] \arrow[d] & B \arrow[d] \\ A \arrow[r] & A+B \end{tikzcd}$$
is a pullback